Chemistry • Atomic number, electron configuration and trends

Periodic Table and Periodicity

Official exercise • Complete solved answers • Original trend diagram

Chapter 8 • Complete official exercise • 10 MCQs + 10 short + 5 constructed + 6 descriptive + 3 investigative

What this chapter is about

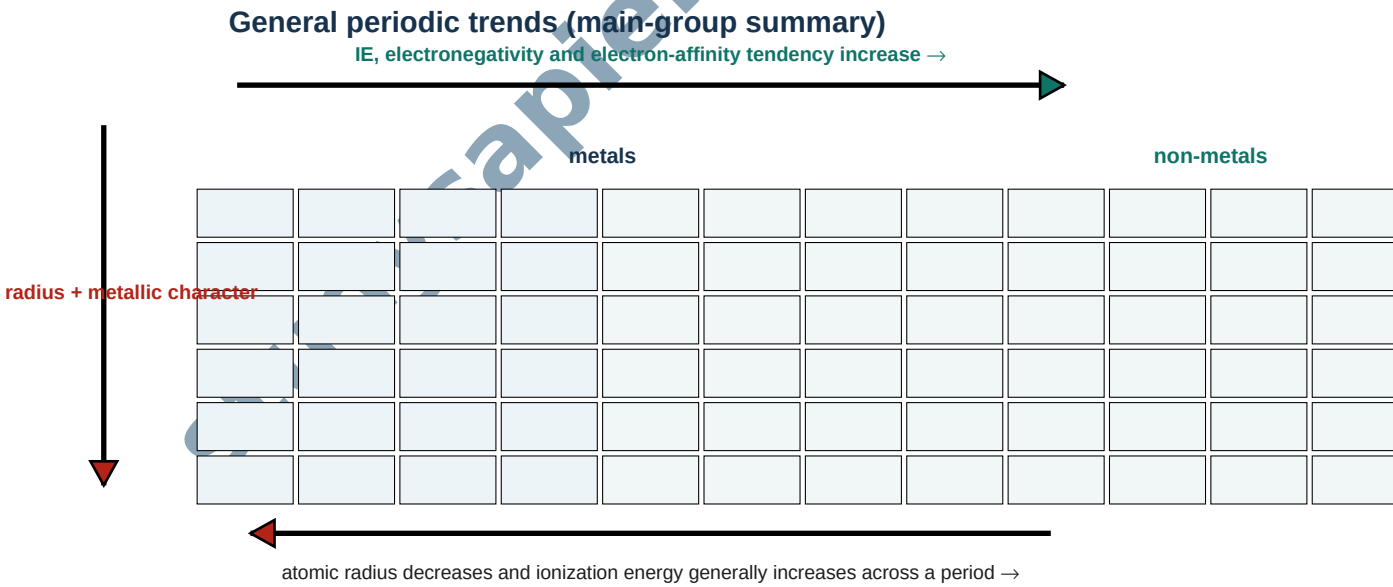
• **The modern periodic law** arranges elements by increasing atomic number. Period number reflects occupied shells; group reflects related valence configuration and properties.

• **Blocks** (s, p, d, f) identify the subshell receiving the differentiating electron and help organize element behaviour.

• **Periodic trends** arise from effective nuclear charge, shielding, atomic size and electron arrangement.

• **Across a period:** radius and metallic character generally decrease while ionization energy and electronegativity generally increase. **Down a group:** the reverse trends dominate.

Figure 1 — Original summary of general periodic trends; individual exceptions are discussed below



Exercise coverage: All five official sections are solved in book order. The transition-metal option is reproduced as printed and clarified in its explanation.

A. Multiple Choice Questions

1. In which period and group will you place the element which is an important part of the solar cell?

(a) third period and 14th group
(b) second period and 14th group
(c) third period and 15th group
(d) third period and 16th group

Answer: (a) third period and 14th group — Silicon is widely used in solar cells. Its configuration [Ne]$3s^2 3p^2$ places it in period 3, group 14.

2. Identify the electronic configuration of the outermost shell of a transition metal.

(a) ns^2np^4
(b) nd^xns^2
(c) ns^2np^6
(d) ns^2np^5

Answer: (b) nd^xns^2 — Option (b) identifies the d-block pattern; it is written more precisely as $(n-1)d^xns^{1-2}$, with some exceptions.

3. Which is the softest metal?

(a) Na
(b) Ca
(c) Al
(d) Zn

Answer: (a) Na — Sodium is a soft alkali metal that can be cut with a knife; the other listed metals are harder.

4. A yellow solid element exists in allotropic forms and is also present in fossil fuels. Indicate the name.

(a) carbon
(b) iodine
(c) aluminium
(d) sulphur

Answer: (d) sulphur — Sulphur is yellow, has rhombic and monoclinic allotropes and occurs in fossil fuels.

5. How many electrons can nitrogen accept in its outermost shell?

(a) 2
(b) 3
(c) 4
(d) 5

Answer: (b) 3 — Nitrogen has five valence electrons and needs three more to complete its octet, forming N^{3-} in suitable ionic compounds.

6. Which element is the most reactive element?

(a) oxygen
(b) chlorine
(c) fluorine
(d) nitrogen

Answer: (c) fluorine — Fluorine is the most reactive listed non-metal and the most electronegative element.

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7. Which element has the highest melting point?

(a) Na (b) K
(c) Rb (d) Cs

Answer: (a) Na — Among these alkali metals, melting point decreases down the group, so sodium has the highest value (about 98°C).

8. In what order does metallic character change in the second group?

(a) Mg > Ca > Ba > Sr (b) Sr > Ba > Ca > Mg
(c) Mg > Sr > Ca > Ba (d) Ba > Sr > Ca > Mg

Answer: (d) Ba > Sr > Ca > Mg — Metallic character increases down group 2: Ba > Sr > Ca > Mg.

9. Which order best describes the atomic radii of oxygen, fluorine and nitrogen?

(a) O < F < N (b) N < F < O
(c) F < O < N (d) O < N < F

Answer: (c) F < O < N — Across a period, increasing effective nuclear charge contracts the atoms, so radius F < O < N.

10. An element having a low ionization energy and low electron affinity is likely to belong to:

(a) group 1 (b) group 13
(c) group 16 (d) group 17

Answer: (a) group 1 — Group 1 metals lose an electron readily and have little tendency to gain another, so both values are comparatively low.

B. Questions for Short Answers

8.1. Why was atomic number chosen to arrange the elements in the periodic table?

Atomic number Z is the number of protons, a unique whole-number property of each element. It determines the electron configuration of a neutral atom and therefore its chemical properties. Ordering by Z also places isotopes of an element together and resolves several mass-order anomalies, such as argon/potassium and tellurium/iodine.

8.2. What is the significance of the word periodic?

“Periodic” means recurring at regular intervals. Elements with similar valence-electron arrangements recur as atomic number increases, so properties such as valency, metallic character and many physical characteristics show repeating trends.

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8.3. Why does the size of a period increase as we move down the periodic table?

Lower periods fill only s and p subshells, whereas higher periods can also fill d and f subshells. More available orbitals allow more elements before the next noble-gas configuration is reached. The resulting period lengths are 2, 8, 8, 18, 18, 32 and 32; thus period length increases in steps rather than strictly at every period.

8.4. In a group, elements have the same number of electrons in the outermost shell. Why is it so?

A new period begins after a noble-gas-like completed outer arrangement. Elements in the same main-group column build their new shells with the same valence pattern, for example ns^1 in group 1 and ns^2np^2 in group 14. Their total inner electrons differ, but their outer configuration recurs.

8.5. Do you expect calcium to be more reactive than sodium? Give the reason.

No. Under the usual metal-reactivity comparison, **sodium is more reactive than calcium**. Sodium loses its single valence electron more easily and has a lower first ionization energy. Calcium must lose two electrons to form Ca^{2+} and has stronger metallic bonding and a higher ionization requirement.

8.6. Which element has the maximum atomic radius and which has the minimum atomic radius in the third period?

Using the textbook's left-to-right radius trend, **sodium (Na) has the maximum and argon (Ar) the minimum** radius in period 3.

Measurement note: Noble-gas radii are not directly comparable with ordinary covalent radii. If only comparable bonded/covalent radii are tabulated, chlorine is often quoted as the smallest third-period atom because argon has no ordinary covalent bond radius. The expected trend-based answer is Na and Ar.

8.7. Why are the most electronegative elements present in the sixteenth and seventeenth groups?

These groups lie near the right of the table, where atoms have high effective nuclear charge, relatively small radii and nearly complete valence shells. They therefore attract shared electrons strongly. Oxygen and sulphur belong to group 16, and fluorine and chlorine to group 17; nitrogen in group 15 is also highly electronegative.

8.8. The first ionization energy of magnesium is less than the second. Give reason.

The first electron is removed from a neutral Mg atom, while the second is removed from Mg^+, which already has a net positive charge and a smaller electron cloud. The remaining electron is attracted more strongly, so more energy is required: $Mg \rightarrow Mg^+ + e^-$ (about 737 kJ mol^{-1}), then $Mg^+ \rightarrow Mg^{2+} + e^-$ (about 1450 kJ mol^{-1}).

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8.9. Is it possible for two metals or two non-metals to form an ionic bond?

In the simple elemental case intended, **no**. Two metal atoms have relatively low ionization energies and tend to bond metallically rather than transfer electrons to one another. Two non-metal atoms have similar tendencies to attract electrons and usually share them covalently.

Precision: Compounds containing only non-metallic elements can nevertheless be ionic when they contain polyatomic ions, for example NH_4NO_3 (NH_4^+ and NO_3^-). Thus “only non-metals can never form ionic compounds” would be too broad.

8.10. Which element has the least ionization energy and which has the highest electronegativity?

In the usual school periodic table, **caesium (Cs)** is cited as having the lowest ionization energy among naturally occurring representative elements, and **fluorine (F)** has the highest electronegativity (Pauling value about 4.0). Francium is sometimes predicted to have a slightly lower ionization energy, but Cs is the standard textbook answer.

C. Constructed Response Questions

8.1. Suppose a new element is discovered. Where would you accommodate it in the periodic table?

First establish its **atomic number Z**, which fixes its exact sequence. Then determine its ground-state electron configuration:

• **Period:** the highest occupied principal shell n for a main-group element.

• **Group:** use its valence configuration and the appropriate main-group or transition-metal column.

• **Block:** identify the subshell receiving the differentiating electron—s, p, d or f.

Predicted physical and chemical properties can confirm the placement, but atomic number and electronic structure are the fundamental criteria.

8.2. What is the first element of the periodic table? Will it lose an electron or gain it?

The first element is **hydrogen, H (Z = 1)**, with configuration $1s^1$. It can lose its electron to form H^+, gain one to form H^- (a hydride with very active metals), or share it covalently. Its behaviour therefore depends on the reaction partner; simple proton loss and covalent sharing are especially common.

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8.3. Atomic radii of boron and aluminium are 88 pm and 125 pm respectively. Which element is expected to lose electrons easily?

Aluminium loses electrons more easily. Its larger radius means its outer electrons are farther from the nucleus and experience more shielding and weaker attraction, giving it a lower ionization energy than boron.

8.4. How would you find the atomic radius of an atom?

An atom has no sharply defined outer edge, so an operational radius is measured. For a covalent radius, determine the internuclear distance between two identical bonded atoms using diffraction or spectroscopic data and take half:

$$r = d(\text{nucleus-to-nucleus}) / 2$$

Example: if d(C–C) = 154 pm, then r(C) = 77 pm

Ionic, metallic and van der Waals radii use different reference situations, so values obtained by different definitions should not be mixed without stating the convention.

8.5. Why is it not possible for an oxygen atom to accept three electrons to form O^{3-} like nitrogen can form N^{3-}?

Oxygen has six valence electrons and needs only two to reach the stable neon configuration. O^{2-} therefore has a complete n = 2 octet. Adding a third electron would require a higher-energy n = 3 orbital and would face strong repulsion in an already negative ion, so O^{3-} is not stable under ordinary chemical conditions.

Nitrogen has five valence electrons and needs three to complete its n = 2 octet, so N^{3-} can form in suitable ionic nitrides. Even there, lattice energy from surrounding cations is important for stabilizing the highly charged ion.

D. Descriptive Questions

8.1. Which information is needed to locate an element in the periodic table if you do not know its atomic number? Is atomic mass helpful?

The most useful information is its **electron configuration**, especially the highest occupied shell, valence-electron pattern and differentiating subshell. Chemical family, common oxidation states and properties can support the assignment.

Period ← occupied shells; group ← valence pattern; block ← differentiating subshell

Atomic mass is not a reliable locating criterion. It is an abundance-weighted average of isotopes and does not uniquely identify an element. Several elements have similar masses, and historical mass ordering produces inversions such as Ar/K and Te/I. Atomic number, not atomic mass, is the fundamental ordering key.

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8.2. How many blocks of elements are present in the periodic table? Are these blocks helpful in studying properties?

There are **four blocks: s, p, d and f**, named according to the subshell receiving the differentiating electron.

Block	Typical location / members	Property guidance
s	Groups 1–2 and helium	Active metals (except H); characteristic valence and ion formation.
p	Groups 13–18	Metals, metalloids and non-metals; varied covalent and ionic behaviour.
d	Transition series	Metals with variable oxidation states, coloured ions and catalytic behaviour.
f	Lanthanides and actinides	Inner-transition metals with characteristic f-electron chemistry.

Blocks help predict electron configuration, valency, metallic character and bonding, although individual exceptions still require the actual configuration.

8.3. Explain the variation in periods of (a) atomic radius and (b) ionization energy, giving reasons.

(a) Atomic radius

Across a period, electrons enter the same principal shell while proton number increases. Shielding does not increase proportionately, so effective nuclear charge rises and pulls the electron cloud closer. Atomic radius therefore **generally decreases** left to right.

(b) Ionization energy

The stronger effective attraction and smaller radius make an outer electron harder to remove, so first ionization energy **generally increases** left to right.

There are local exceptions caused by subshell stability and electron repulsion. For example, Be has higher first ionization energy than B because B's new 2p electron is easier to remove, and N has a higher value than O because repulsion in O's paired 2p orbital lowers its ionization energy.

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8.4. Which physical properties of elements may lead us to know what type of bond it will form?

Observed property	Bonding / structure suggested
Metallic lustre, malleability and good solid-state conductivity	Metallic bonding and delocalized electrons
High-melting brittle crystals, conducting when molten or dissolved	Ionic lattice
Low melting/boiling point, soft molecular solid, liquid or gas; non-conductor	Simple covalent molecules with weak intermolecular forces
Very hard, very high-melting non-conductor	Giant covalent network, such as diamond or SiO_2
Water/organic-solvent solubility behaviour	May support ionic/polar versus non-polar covalent character, but is not conclusive alone

No single property is proof in every case. Conductivity conditions, solubility, structure and chemical behaviour should be considered together; graphite is a notable covalent conductor.

8.5. Write down the names of four non-metals which exist in the solid state at normal temperature.

Four examples are **carbon (C), sulphur (S), phosphorus (P) and iodine (I)**. Selenium is another solid non-metal.

8.6. Why do the second and third periods have equal numbers of elements while all other periods contain different numbers?

Both periods contain eight elements because each fills one s and three p orbitals: 2s2p in period 2 and 3s3p in period 3. These eight orbitals accommodate a total of eight electrons, hence eight elements.

Although 3d exists, its filling begins only after 4s in period 4 because of orbital-energy ordering. Therefore period 3 does not include the ten 3d elements.

Period lengths: 2, 8, 8, 18, 18, 32, 32

Longer periods include d or f subshells, giving different total element counts.

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E. Investigative Questions

8.1. The arrangement of elements as a periodic table is a remarkable achievement. Comment on its benefits.

The periodic table converts a large collection of unrelated facts into an organized predictive system.

• **Classification:** It groups elements with related valence configurations and chemical behaviour.

• **Prediction:** It allows estimation of radius, ionization energy, electronegativity, metallic character, valency and likely bond type.

• **Relationships:** Periods, groups and blocks reveal similarities, gradual changes and diagonal relationships.

• **Discovery:** Mendeleev's gaps guided predictions of undiscovered elements; the modern table also accommodates synthetic elements.

• **Learning and research:** It provides a compact framework for formulas, reactions, materials design and comparison of new data.

Its greatest strength is that the pattern follows **atomic number and electronic structure**, so it explains properties rather than merely listing them.

8.2. Both lithium and beryllium behave differently from the other alkali and alkaline-earth metals. Suggest reasons.

Li and Be are the first members of their groups and have exceptionally small atomic and ionic radii, high charge density, relatively high ionization energies and greater polarizing power. They have no d orbitals in their valence shell and only the n = 2 valence region.

These features give their compounds more covalent character and produce diagonal similarities: Li resembles Mg in several respects, while Be resembles Al.

• Li forms Li_2O rather than the peroxides/superoxides common among heavier alkali metals, and it reacts directly with N_2 to form Li_3N.

• Be compounds such as $BeCl_2$ have substantial covalent character; $Be(OH)_2$ is amphoteric, unlike the mainly basic hydroxides of heavier group 2 metals.

High hydration energy of the tiny Li^+ ion also strongly influences lithium's solution chemistry.

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8.3. The modern periodic table is an amended form of Mendeleev's earlier table. How are they different?

Feature	Mendeleev table	Modern periodic table
Main ordering property	Increasing atomic mass	Increasing atomic number Z
Theoretical basis	Empirical periodic recurrence of properties	Electron configurations and quantum subshells
Isotopes	Not understood or assigned a clear common place	All isotopes occupy the same element position
Noble gases	Absent because they had not yet been discovered	Included as group 18
Unknown elements	Gaps left and properties predicted	Atomic-number positions clarify placement of natural and synthetic elements
Structure	Fewer groups and a less complete layout	18 groups, 7 periods and s, p, d, f blocks
Mass-order anomalies	Ar/K, Co/Ni and Te/I create difficulties	Ordering by Z resolves them

The modern table preserves Mendeleev's central insight—periodic recurrence—but replaces atomic mass with the more fundamental atomic number.

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Points to Remember

Modern periodic law: properties are periodic functions of atomic number.

There are 18 groups, 7 periods and four blocks: s, p, d and f.

Across a period, effective nuclear charge generally increases; radius decreases, while ionization energy and electronegativity generally increase.

Down a group, added shells and shielding increase radius and metallic character while ionization energy generally decreases.

Group similarity comes mainly from recurring valence-electron configuration.

First ionization removes an electron from a neutral gaseous atom; successive ionizations require more energy.

Atomic-radius values depend on definition: covalent, metallic, ionic and van der Waals radii are not interchangeable.

Atomic mass does not uniquely locate an element; atomic number and electron configuration do.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	a	2	b	3	a	4	d	5	b
6	c	7	a	8	d	9	c	10	a

How to attempt this chapter in the paper

• Before applying a trend, identify the direction: across a period or down a group, then give effective nuclear charge or shielding as the reason.
• Learn general trends but mention named subshell exceptions when a question supplies actual ionization data.
• Use Z for position and A only for isotope/nucleon information; never use atomic mass as a unique identifier.
• For bond-type predictions, combine more than one property and state the conditions for conductivity.